

A COMPUTATIONAL BAYESIAN APPROACH TO THE BALANCED BÜHLMANN CREDIBILITY MODEL

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Summary: Credibility theory has long been used in the insurance industry and actuarial science as part of the process of estimating risks and setting premiums. A fully Bayesian analysis has, however, proved to be computationally infeasible. The reason for this is that the integration operation plays a fundamental role in Bayesian statistics.

In this note, simple numerical methods to obtain exact posterior distributions and predictive densities for the balanced Bühlmann model are described. These distributions are obtained through Monte Carlo simulations that generate independent samples from the joint posterior distribution. These methods are therefore preferable to methods such as Gibbs sampling, which generates dependent samples from the joint distribution. The methods discussed can also be extended to more complicated credibility models.

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1. Introduction

Statistical methods with a Bayesian flavour, in particular credibility theory, have long been used in the insurance industry and actuarial science as part of the process of estimating risks and setting premiums. The Bayesian methodology is, however, computer intensive. Multi-dimensional integration is required in many situations. The difficulty in obtaining distributions analytically, increases with the complexity of the model as well as with the dimension of the parameter space. The technical problems encountered in attempting to carry out the required numerical integrations for obtaining the marginal posterior and predictive densities, have long served as an impediment to the wider application of the Bayesian framework to real data and credibility models.

The fast evolution of computing power and numerical integration techniques such as Markov Chain Monte Carlo methods and, more specifically, Gibbs sampling for evaluating high-dimensional posterior integrals now facilitates the use of a full analysis.

In this note, simple numerical methods to obtain exact posterior distributions and predictive densities for the balanced Bühlmann credibility model are described. These distributions are obtained through Monte Carlo simulations that generate independent samples from the joint posterior distribution. The implementation of the methods is therefore easier than Gibbs sampling, which generates dependent samples from the joint distribution. In addition, the monitoring of stochastic convergence and mixing of the Markov chain in the case of Gibbs sampling can so far only be assessed with convergence diagnostics.

Klugman (1992) states:

“Over the years, the word ‘credibility’ has come to mean different things to different people, usually based on the restrictions that are placed on the solution. What all credibility problems have in common though, is the problem to be solved:

Based on observations from both members of the class under consideration and members of the other classes, estimate the distribution (or its properties) of future claims from members of the class under consideration.”

Miller and Hickman (1975) gave the following definition of credibility theory:

“Credibility theory may be thought of as involving the development of a system of weights, which are functions of the size of the insurance experience available; for estimating future claim-costs as a weighted average of the actual claims for the group under consideration and the expected claims based on prior or ancillary experience.”

Credibility theory originated with the proposal by Whitney (1918) of a premium that represents a weighted average between a firm’s individual claim experience and that of the industry as a whole. Charging a premium based on individual as well as collective experience is justified according to Dannenburg, Kaas and Goovaerts (1996) as well as Klugman (1992), since:

- (i) the portfolio is neither completely homogeneous, nor completely heterogeneous,
- (ii) insured individuals or groups will seldom provide enough information to estimate their potential for future claims accurately,
- (iii) insured individuals or groups will not be penalized to the full extent (or rewarded with a zero premium) indicated by their experience.

Klugman (1992) further remarks that the difficult problem is the determination of the relative emphasis to place on the specific and overall

experiences, and continue to state that a Bayesian approach is the best way to resolve this.

The compromise between the specific and overall experiences, i.e. the weighted average, can be denoted by

$$\tilde{Z}\bar{Y}_i + (1 - \tilde{Z})\bar{Y}_{..}$$

where $\bar{Y}_i$ is, for example, the average claim for the i^{th} company ($i = 1, \dots, I$) and $\bar{Y}_{..}$ is the average claim over all the companies. $\tilde{Z}$ is known as the credibility factor and is a number between zero and one. The credibility factor expresses how ‘credible’ the individual experience of insurer i is. A premium in this weighted average form is called a credibility premium. The data are not necessarily limited to claim figures. One might as well consider loss ratios, claims as a percentage of sums insured and so forth.

The emphasis of our research will be on the ‘greatest accuracy credibility theory’, in which the credibility factors follow as optimal coefficients in a Bayesian model incorporating variance components. Bühlmann developed this model in the 1960s (Bühlmann, 1967).

For a historical overview of credibility, according to Makov, Smith and Liu (1996), see Hickman (1975) and Miller and Hickman (1975). For an ‘un-Bayesian’ approach to credibility theory see Gerber (1982). More recent reviews of credibility theory are given in a special issue of *Insurance: Abstract and Reviews* (February 1986).

Other reviews are given in Waters (1987), Goovaerts and Hoogstad (1987), Dannenburg, Kaas and Goovaerts (1996), Kaas and Goovaerts (1998), and Goovaerts *et al.* (1990). The most recent and complete review is Norberg (2004). For generalised linear mixed model specifications of well-known credibility models see Antonio and Beirlant (2005).

In section 2, the balanced (classical) model of Bühlmann (1967) is presented to illustrate the ideas of credibility models and in section 3, a Bayesian analysis of the random effects model is considered. It is also shown that the posterior mean of a group or company is equal to the credibility estimator of the balanced Bühlmann model. In section 4, a Monte Carlo simulation procedure is explained to obtain independent samples of the variance components from their joint posterior distribution and in section 5, an illustrative example is provided. In section 6 the sampling-importance resampling (SIR) idea is examined that provides a technique whereby independent samples from one distribution may be modified to form samples from another distribution and is thus preferable to methods such as Gibbs sampling, which generates dependent samples from the joint distribution.

2. The balanced Bühlmann model

As mentioned by Makov, Smith and Liu (1996), in the case of the balanced Bühlmann model, the observations are decomposed into independent variance components and it is balanced in the sense that the number of observation-years for each contract and the variance of each observation is the same. The risk parameter is homogeneous in time and the i^{th} company is described by the vector $[\theta_i \ \underline{Y}'_i]' \equiv [\theta_i \ Y_{i1} \ Y_{i2} \dots Y_{iJ}]'$. Here both the vectors $[\theta_i \ \underline{Y}'_i]'$ and the conditional variables $Y_{ij}|\theta_i$ ($i = 1, \dots, I$; $j=1, \dots, J$) are assumed independent and identically distributed, where Y_{ij} is the j^{th} claim for the i^{th} company and the parameter θ_i is the expected claim for the i^{th} company. In the next paragraph it will be shown that the mean of the posterior distribution of θ_i , in the case of the balanced random effects model, is equal to the credibility

estimator of the balanced Bühlmann model, viz.

$$\hat{Z}\bar{Y}_{i.} + (1 - \hat{Z})\bar{Y}_{..} \quad (2.1)$$

where $\bar{Y}_{i.}$ is the average claim for the i^{th} company ($i = 1, \dots, I$) and $\bar{Y}_{..}$ is the average claim over all the companies obtained from the data. $\bar{Z}$ is known as the credibility factor and is a number between zero and one.

3. Random effects model

To illustrate how random effects models or variance component models arise, consider an example of an insurance process, which produces claims, and where the companies were taken at random. On the assumption that the companies and claims vary independently, and additively contribute errors (residuals) r_i and ε_{ij} , we have the mathematical model

$$Y_{ij} = \theta + r_i + \varepsilon_{ij}; \text{ for } i = 1, \dots, I \text{ and } j = 1, \dots, J. \quad (3.1)$$

The random variables r_i and ε_{ij} are sometimes called random effects and the model in (3.1) is known as a balanced random effects model. Furthermore, we assume that

$$\varepsilon_{ij} \sim N(0, \sigma_1^2) \text{ and } r_i \sim N(0, \sigma_2^2).$$

In addition, the single fixed effect θ denotes the overall mean claim and the random effect r_i denotes the deviation from this mean, specific to company i . According to Dannenburg, Kaas and Goovaerts (1996) and Kaas and Goovaerts (1998), this model is a simplified form of the classical Bühlmann model. One simplification is that independence and normality of the errors are assumed, while Bühlmann only postulates zero correlation. Additionally, not only do all the observations in this model have the same variance, but all the cells also contain the same number of observations. It is for these reasons that

Dannenburg, Kaas and Goovaerts (1996) refer to it as the balanced Bühlmann credibility model.

Equation (3.1) can also be written as $Y_{ij} = \theta_i + \varepsilon_{ij}$. As prior, we will use the noninformative prior as defined by Box and Tiao (1973), page 251, viz.

$$p(\theta, \sigma_1^2, \sigma_2^2) \propto \sigma_1^{-2} (\sigma_1^2 + J\sigma_2^2)^{-1}. \quad (3.2)$$

This prior may easily be obtained by applying Jeffreys' rule. Jeffreys' rule states that the prior distribution for a set of parameters is taken to be proportional to the square root of the determinant of the Fisher information matrix. Berger and Bernardo (1992) proved that it is also a reference prior for the group orderings $\{\sigma_1^2, \theta, \sigma_2^2\}$, $\{\sigma_1^2, \sigma_2^2, \theta\}$ and $\{\theta, \sigma_1^2, \sigma_2^2\}$. In Theorem 3.1 it will be proved that (3.2) is also a probability matching prior with respect to the error variance σ_1^2 .

Theorem 3.1 For the balanced Bühlmann model defined in equation (3.1), the probability matching prior is given by equation (3.2).

The proof is given in the appendix.

By using the results in Box and Tiao (1973), it can easily be shown that the joint posterior distribution of the variance components is given by

$$p(\sigma_1^2, \sigma_2^2 | \underline{Y}) \propto (\sigma_1^2)^{-\frac{1}{2}(\nu_1+2)} (\sigma_1^2 + J\sigma_2^2)^{-\frac{1}{2}(\nu_2+2)} \exp \left\{ -\frac{1}{2} \left[\frac{\nu_1 m_1}{\sigma_1^2} + \frac{\nu_2 m_2}{\sigma_1^2 + J\sigma_2^2} \right] \right\} \quad (3.3)$$

where $\sigma_1^2 > 0$ and $\sigma_2^2 > 0$; and also that the posterior distribution of θ , given the variance components, is normal with mean

$$E(\theta | \underline{Y}, \sigma_1^2, \sigma_2^2) = \bar{Y}_{..} \quad (3.4)$$

and variance

$$Var(\theta | \sigma_1^2, \sigma_2^2) = \frac{1}{IJ} (\sigma_1^2 + J\sigma_2^2) \quad (3.5)$$

where

$$\begin{aligned} \nu_1 &= I(J-1); \quad \nu_2 = I-1; \quad \bar{Y}_{i.} = \frac{1}{J} \sum_{j=1}^J Y_{ij}; \quad \bar{Y}_{..} = \frac{1}{IJ} \sum_{i=1}^I \sum_{j=1}^J Y_{ij}; \\ \nu_1 m_1 &= \sum_{i=1}^I \sum_{j=1}^J (Y_{ij} - \bar{Y}_{i.})^2; \quad \nu_2 m_2 = J \sum_{i=1}^I (\bar{Y}_{i.} - \bar{Y}_{..})^2; \quad \text{and} \\ \underline{Y} &= [Y_{11} \ Y_{12} \ \dots \ Y_{1J} \ Y_{21} \ \dots \ Y_{IJ}]'. \end{aligned}$$

The proof of the following theorem is given in Van der Merwe and Bekker (2004).

Theorem 3.2

- (i) The posterior distribution of $\theta_i = \theta + r_i$, given the variance components, is normal with mean

$$E(\theta_i | \underline{Y}, \sigma_1^2, \sigma_2^2) = \frac{J\sigma_2^2}{\sigma_1^2 + J\sigma_2^2} \bar{Y}_{i.} + \frac{\sigma_1^2}{\sigma_1^2 + J\sigma_2^2} \bar{Y}_{..}; \quad i = 1, \dots, I \quad (3.6)$$

and variance

$$Var(\theta_i | \underline{Y}, \sigma_1^2, \sigma_2^2) = \frac{\sigma_1^2}{\sigma_1^2 + J\sigma_2^2} \left(\frac{\sigma_1^2}{IJ} + \sigma_2^2 \right). \quad (3.7)$$

- (ii) The predictive probability density function of a vector of q future observations from the i^{th} group (q future claims from the i^{th} company), given the variance components, is multivariate normal with mean vector

$$E(\tilde{\underline{Y}}_i | \underline{Y}, \sigma_1^2, \sigma_2^2) = \frac{J\sigma_2^2}{\sigma_1^2 + J\sigma_2^2} \bar{Y}_{i.} \underline{1} + \frac{\sigma_1^2}{\sigma_1^2 + J\sigma_2^2} \bar{Y}_{..} \underline{1} \quad (3.8)$$

and covariance matrix

$$Var(\tilde{\underline{Y}}_i | \underline{Y}, \sigma_1^2, \sigma_2^2) = \frac{\sigma_1^2}{\sigma_1^2 + J\sigma_2^2} \left(\frac{\sigma_1^2}{IJ} + \sigma_2^2 \right) \underline{1}\underline{1}' + \sigma_1^2 \tilde{I} \quad (3.9)$$

where $\tilde{I}(q \times q)$ is the identity matrix; $\underline{1}(q \times 1)$ is a vector of ones; and $\tilde{\underline{Y}}_i(q \times 1) = [Y_{i,J+1} \ Y_{i,J+2} \ \dots \ Y_{i,J+q}]' = [\tilde{Y}_{i,1} \ \tilde{Y}_{i,2} \ \dots \ \tilde{Y}_{i,q}]'$ is a vector of future observations from the i^{th} group.

It is worth pointing out that the q elements in the mean vector (3.8) are identical, as are all the variances in the covariance matrix (3.9).

This also applies to the covariances.

- (iii) The predictive probability density function of a vector of q

future observations from a new or unknown group (company),
 $\underline{Y}^* = [Y_1^* \ Y_2^* \ \dots \ Y_q^*]'$, given the variance components σ_1^2 and σ_2^2 , is normal with mean vector

$$E(\underline{Y}^* | \underline{Y}, \sigma_1^2, \sigma_2^2) = \bar{Y}_{..} \underline{1} \quad (3.10)$$

and covariance matrix

$$Var(\underline{Y}^* | \underline{Y}, \sigma_1^2, \sigma_2^2) = \sigma_1^2 \tilde{I} + \sigma_2^2 \underline{1}\underline{1}' + \frac{1}{IJ} (\sigma_1^2 + J\sigma_2^2) \underline{1}\underline{1}' \quad (3.11)$$

Corollary

- (i) From equations (3.8) and (3.9) it follows that $\bar{Y}_i = \frac{1}{q} \sum_{l=1}^q \tilde{Y}_{il}$ the mean of q future claims from the i^{th} company, is normally distributed with mean

$$E(\bar{Y}_i | \underline{Y}, \sigma_1^2, \sigma_2^2) = \frac{J\sigma_2^2}{\sigma_1^2 + J\sigma_2^2} \bar{Y}_{i.} + \frac{\sigma_1^2}{\sigma_1^2 + J\sigma_2^2} \bar{Y}_{..} \quad (3.12)$$

and variance

$$Var(\bar{Y}_i | \underline{Y}, \sigma_1^2, \sigma_2^2) = \frac{\sigma_1^2}{\sigma_1^2 + J\sigma_2^2} \left(\frac{\sigma_1^2}{IJ} + \sigma_2^2 \right) + \frac{1}{q} \sigma_1^2 \quad (3.13)$$

- (ii) It is also clear from (3.10) and (3.11) that $\bar{Y}^* = \frac{1}{q} \sum_{j=1}^q Y_j^*$, the mean of q future claims from a new or unknown company is normally distributed with mean

$$E(\bar{Y}^* | \underline{Y}, \sigma_1^2, \sigma_2^2) = \bar{Y}_{..} \quad (3.14)$$

and variance

$$Var(\bar{Y}^* | \underline{Y}, \sigma_1^2, \sigma_2^2) = \frac{1}{q} \sigma_1^2 + \sigma_2^2 + \frac{1}{IJ} (\sigma_1^2 + J\sigma_2^2). \quad (3.15)$$

From equation (3.6), it can also be seen that

$$E(\theta_i | \underline{Y}, \sigma_1^2, \sigma_2^2) = \tilde{Z} \bar{Y}_{i.} + (1 - \tilde{Z}) \bar{Y}_{..}$$

where $\tilde{Z} = \frac{J\sigma_2^2}{\sigma_1^2 + J\sigma_2^2}$, i.e. the mean of the posterior distribution of θ_i , given the variance components, is equal to the credibility estimator of the balanced Bühlmann model. This is also true for the mean of the predictive density of future claims from the i^{th} company.

4. Monte Carlo simulation

The usual approach to the problem of predicting linear combinations of fixed and random effects $(\theta + r_i, \quad i = 1, \dots, I)$, when the values of the variance components are not known, has been to estimate the variance components and to proceed thereafter as if these estimates were the true values.

Patterson and Thompson (1971) developed a method to derive unbiased estimates of the unknown variance components σ_1^2 and σ_2^2 based on the maximum likelihood principle; it is called restricted maximum likelihood estimation (REML). Substituting the REML estimates of σ_1^2 and σ_2^2 yields the empirical Bayes solution to the random effects model. For an example on this, see Chapter 8 on Credibility Theory in the ActEd 106 Actuarial Study Guide, 2002. In our opinion, there are several problems with this approach:

- (i) The properties of the predictors are hard to assess.
- (ii) When the values of the variance components are estimated from the data, their sampling errors are generally not taken into account in the subsequent analysis. Therefore, the variance of the prediction error will generally be underestimated.
- (iii) Depending upon the size and characteristics of the data, point estimators of variance components can be highly variable. For certain values of the variance component estimators, the predictors, which are obtained by substituting these values in the “Best Linear Unbiased Predictor”, are intuitively unappealing.

An alternative approach to the empirical Bayesian approach is the fully Bayesian approach.

In many Bayesian problems, marginal posterior distributions are needed to make appropriate inferences. Technical difficulties that arise in the calculation of the marginal posterior densities needed in Bayesian inference have long served as an impediment to the wider application of the Bayesian framework to real data. The reason for this is that the integration operation plays a

fundamental role in Bayesian statistics. In the last few years, there have been a number of advances in the numerical integration and analytic approximation techniques for such calculations, but implementation of these approaches typically requires sophisticated numerical or analytic approximation expertise and, possibly, specialist software.

However, due to work of Gelfand and Smith (1991), Gelfand *et al.* (1990), Carlin *et al.* (1992) and Gelfand *et al.* (1992), Markov Chain Monte Carlo (MCMC) procedures, and more specifically the Gibbs sampler, have been shown as a useful tool for applied Bayesian inference in a broad variety of statistical problems. The Gibbs sampler is implicit in the work of Hastings (1970) and made popular in the image-processing context by Geman and Geman (1984). The Gibbs sampler is an adaptive Monte Carlo integration technique. The typical objective of the sampler is to collect a sufficiently large enough number of parameter realizations from conditional posterior densities in order to obtain accurate estimates of the marginal posterior densities. The principal requirement of the sampler is that all conditional densities must be available in the sense that they can generate random variates.

The Gibbs sampler, as mentioned earlier, is appealing for its general applicability and ease of implementation. However, the burden of proof has shifted to the monitoring of stochastic convergence and mixing of the Markov chain, which so far can only be assessed with convergence diagnostics, see Robert and Casella (1999), as well as Jones and Hobert (2001). As pointed out by Gelfand (2002), “in general, convergence can never be assessed, as comparison can be made only between different iterations of one chain or between different observed chains, but never with the true stationary distribution”. Because of this problem, intense research has also focused on generating samples distributed perfectly as the stationary distribution of the

Markov chain, see Green and Murdoch (1999), as well as Casella, Lavine and Robert (2001). Unfortunately, such an algorithm is currently feasible for only limited low-dimensional problems, and the cost of obtaining $\tilde{l}$ samples is far greater than that of the usual MCMC, because, essentially, the entire algorithm must be repeated $\tilde{l}$ times.

In this paragraph, two simple numerical methods to obtain exact posterior distributions and predictive densities for the balanced Bühlmann model will be described. These distributions are obtained through Monte Carlo simulations where independent samples from the joint posterior distribution are obtained. These methods are thus preferable to methods such as Gibbs sampling which generate dependent samples from the joint posterior distribution. In addition, the conditional posterior distribution of σ_2^2 , in the case of the Jeffreys' prior, is not a standard density to simulate from and Metropolis-Hastings with Gibbs sampling should be used. The first method can be explained as follows.

The joint posterior distribution for the variance components is defined in equation (3.3). Let $\sigma_{12}^2 = \sigma_1^2 + J\sigma_2^2$, then $\sigma_{12}^2 > \sigma_1^2$. We find that if this restriction did not apply, then the posterior distributions for σ_1^2 and σ_{12}^2 would be independent; each proportional to an inverse gamma distribution, i.e.

$$p(\sigma_1^2 | \underline{Y}) \propto (\sigma_1^2)^{-\frac{1}{2}(\nu_1+2)} \exp \left\{ -\frac{1}{2} \frac{\nu_1 m_1}{\sigma_1^2} \right\} \quad (4.1)$$

and

$$p(\sigma_{12}^2 | \underline{Y}) \propto (\sigma_{12}^2)^{-\frac{1}{2}(\nu_2+2)} \exp \left\{ -\frac{1}{2} \frac{\nu_2 m_2}{\sigma_{12}^2} \right\} \quad (4.2)$$

and the joint posterior distribution for σ_1^2 and σ_{12}^2 would be the product of these two distributions. However, the restrictions do apply. Nevertheless, using a two-step rejection sampling procedure, it is now straightforward to generate samples from the joint distribution:

- (a) Generate values from the two inverse gamma distributions, and

- (b) Retain those sets of values that conform to the restricted parameter space, $\sigma_{12}^2 > \sigma_1^2$.

We can explain (a) and (b) in a simpler way as follows:

- (i) Simulate τ from a $\chi_{\nu_1}^2$ distribution and calculate $\sigma_1^2 = \frac{\nu_1 m_1}{\tau}$.
 Since $\chi_{\nu_1}^2 = \sum_{i=1}^{\nu_1} Z_i^2$, where $Z_i \sim N(0, 1)$, it is easy to simulate τ .
- (ii) Simulate δ from a $\chi_{\nu_2}^2$ distribution and calculate $\sigma_1^2 + J\sigma_2^2 = \sigma_{12}^2 = \frac{\nu_2 m_2}{\delta}$.
- (iii) Calculate $\sigma_2^2 = \frac{\sigma_{12}^2 - \sigma_1^2}{J}$. It must be mentioned that even though σ_1^2 and σ_{12}^2 will always be positive, the estimated value for σ_2^2 can be negative. If a negative value for σ_2^2 is obtained, disregard this value as well as the corresponding value for σ_1^2 .
- (iv) Repeat steps (i), (ii) and (iii) until $\tilde{l}$ permissible values are obtained. For our example, $\tilde{l}$ was taken as 10 000. By substituting each pair of simulated values (σ_1^2, σ_2^2) in the conditional densities $p(\theta_i | \underline{Y}, \sigma_1^2, \sigma_2^2)$, $p(\tilde{Y}_i | \underline{Y}, \sigma_1^2, \sigma_2^2)$ and $p(\bar{Y}^* | \underline{Y}, \sigma_1^2, \sigma_2^2)$, and by using a Rao-Blackwell argument, see Gelfand and Smith (1991), density estimates of the marginal distributions $p(\theta_i | \underline{Y})$, $p(\tilde{Y}_i | \underline{Y})$ and $p(\bar{Y}^* | \underline{Y})$ can be obtained by averaging the conditional densities over the $\tilde{l}$ repetitions.

For the second method, the transformation $\sigma_1^2 = \sigma_1^2$ and $\tilde{F} = \left(\frac{m_2}{m_1}\right) \left(1 + \frac{J\sigma_2^2}{\sigma_1^2}\right)^{-1}$ is made.

The joint posterior distribution (3.3) can then be written as

$$p(\sigma_1^2 | \tilde{F}, \underline{Y}) p(\tilde{F} | \underline{Y}), \text{ where}$$

$$p(\sigma_1^2 | \tilde{F}, \underline{Y}) \propto (\sigma_1^2)^{-\frac{1}{2}(IJ+1)} \exp \left\{ -\frac{\nu_1 m_1}{2\sigma_1^2} \left[1 + \frac{\nu_2}{\nu_1} \tilde{F} \right] \right\}, \quad \sigma_1^2 > 0 \quad (4.3)$$

which is an Inverted Gamma distribution, and

$$p(\tilde{F} | \underline{Y}) \propto \tilde{F}^{\frac{1}{2}\nu_2 - 1} \left(1 + \frac{\nu_2}{\nu_1} \tilde{F}\right)^{-\frac{1}{2}(\nu_1 + \nu_2)}, \quad 0 \leq \tilde{F} \leq \frac{m_2}{m_1} \quad (4.4)$$

which is a truncated F-distribution with ν_2 and ν_1 degrees of freedom. Since σ_1^2 can be integrated out, it follows that the posterior distribution of θ_i given $\tilde{F}$ is a Student t-distribution with $IJ - 1$ degrees of freedom, mean

$$E(\theta_i | \tilde{F}, \underline{Y}) = \left(1 - \frac{m_1}{m_2} \tilde{F}\right) \bar{Y}_{i.} + \frac{m_1}{m_2} \tilde{F} \bar{Y}_{..} \quad (4.5)$$

and variance

$$Var(\theta_i | \tilde{F}, \underline{Y}) = \frac{1}{J(IJ - 3)} \nu_1 m_1 \left(1 + \frac{\nu_2}{\nu_1} \tilde{F}\right) \left\{1 + \left(\frac{1}{I} - 1\right) \frac{m_1}{m_2} \tilde{F}\right\}. \quad (4.6)$$

The predictive densities of $\tilde{Y}_i$ and $\bar{Y}^*$ will also be t-distributed with $IJ - 1$ degrees of freedom. The mean of $\tilde{Y}_i$ will be the same as that for θ_i (equation (4.5)) but its variance is

$$Var(\tilde{Y}_i | \tilde{F}, \underline{Y}) = \frac{1}{J(IJ - 3)} \nu_1 m_1 \left(1 + \frac{\nu_2}{\nu_1} \tilde{F}\right) \left\{\frac{J}{q} + 1 + \left(\frac{1}{I} - 1\right) \frac{m_1}{m_2} \tilde{F}\right\}. \quad (4.7)$$

In the case of $\bar{Y}^*$, the mean is

$$E(\bar{Y}^* | \tilde{F}, \underline{Y}) = \bar{Y}_{..} \quad (4.8)$$

and variance

$$Var(\bar{Y}^* | \tilde{F}, \underline{Y}) = \frac{1}{(IJ - 3)} \nu_1 m_1 \left(1 + \frac{\nu_2}{\nu_1} \tilde{F}\right) \left\{\frac{1}{q} - \frac{1}{J} + \frac{1}{J} \frac{m_2}{m_1} \frac{1}{\tilde{F}} \left(1 + \frac{1}{I}\right)\right\}. \quad (4.9)$$

The simulation procedure to obtain $\tilde{F}$ (truncated F_{ν_2, ν_1} values) can be explained as follows:

- (i) Generate values from two independent Chi-square distributions and calculate

$$\tilde{F} = \frac{\chi_{\nu_2}^2 / \nu_2}{\chi_{\nu_1}^2 / \nu_1}.$$

- (ii) If $0 \leq \tilde{F} \leq \frac{m_2}{m_1}$, substitute $\tilde{F}$ in the suitable t-distribution but disregard it if it does not conform to the restricted parameter space. Density estimates of the marginal (unconditional) distributions can be obtained by averaging the conditional t-densities over the $\tilde{l}$ repetitions.

There is not much enabling one to choose between the two procedures and they will give the same unconditional posterior and predictive densities. However, we prefer Method 1 because it can easily be extended to Jewell's Hierarchical model and Crossed Classification models.

For details about similar approaches see Wolfinger (1998) and Guttman and Menzefricke (2003).

Klugman (1992) attests that "[t]he two greatest advantages to the actuary of the Bayesian approach are that the method is independent of the model and that interval estimates are as easy to obtain as point estimates. The former attribute means that once one learns how to analyze one problem, the solution to similar, but more complex, problems will be no more difficult. The second one takes on added significance as the actuary of today is expected to provide evidence concerning the quality of any estimates."

As mentioned, the illustrated Bayesian simulation methods are preferable to Gibbs sampling, since they generate independent samples from the joint posterior distribution.

5. Illustrative example

Example 5.1 We will now apply the Bayesian simulation procedure to the following simple data set given on page 46 of Chapter 8 on Credibility Theory in the ActEd 106 Actuarial Study Guide, 2002. The table below shows the aggregate claim amounts (in £m) for an international insurer's fire portfolio for a 5-year period.

Table 5.1. An International Insurer's Annual Fire Portfolio Aggregate Claims ($\bar{Y}_{ij}$ in £m)

Year (j)	1	2	3	4	5	Mean $\bar{Y}_{i.}$
Country (i)						
1	48	53	42	50	59	50.4
2	64	71	64	73	70	68.4
3	85	54	76	65	90	74.0
4	44	52	69	55	71	58.2
						$\bar{Y}_{..} = 62.75$

Based on the data, the quantities needed for the simulation procedure are:

$$I = 4; \quad J = 5; \quad \nu_1 = I(J-1) = 16; \quad \nu_2 = I - 1 = 3; \quad \nu_1 m_1 = \sum_{i=1}^I \sum_{j=1}^J (Y_{ij} - \bar{Y}_{i.})^2 = 1619.2 \text{ and } \nu_2 m_2 = J \sum_{i=1}^I (\bar{Y}_{i.} - \bar{Y}_{..})^2 = 1658.55.$$

In generating values from the posteriors in (4.1) and (4.2), we simulated 31 198 in order to retain the chosen 30 000 sets of values that met the restriction $\sigma_{12}^2 > \sigma_1^2$. In Figures 5.1 and 5.2, histograms of the posterior distributions of σ_1^2 and σ_2^2 as well as the smooth probability density estimates (Rice, 1995:258-259) are illustrated. The histogram of the credibility factor, $\tilde{Z} = \frac{J\sigma_2^2}{\sigma_1^2 + J\sigma_2^2}$, is illustrated in Figure 5.3. The modes and 95% credibility intervals are also given. In what follows, a credibility interval refers to the Bayes confidence interval of a parameter $(\sigma_1^2, \sigma_2^2, \tilde{Z} \text{ or } \theta_i)$. A prediction interval, on the other hand, is a Bayes confidence interval of a future value or the mean of, say, q future values.

Figure 5.1 displays the variability of σ_1^2 , the within group (country) variance, of the aggregate claim amounts. As is often the case, the posterior distribution of σ_1^2 is markedly skew to the right but not as skew as the posterior distribution of the between-groups variance component. The reason for this is that a reasonable number of degrees of freedom, $\nu_1 = 16$, are associated with the error variance. The 95% credibility interval, which is an indication of uncertainty in the true value of σ_1^2 , seems to be reasonably large but it

has to be remembered that the variance σ_1^2 is a squared entity and that the corresponding interval for the standard deviation σ_1 is (7.48, 15.13), which can easily be interpreted. The posterior distributions of σ_2^2 and $\tilde{Z}$, on the other hand, are very skew and the 95% credibility intervals wide, which is a definite indication of the large uncertainty associated with the between-groups variance component of the aggregate claim amounts. The main reason for this is the small number of countries involved in the experiment.

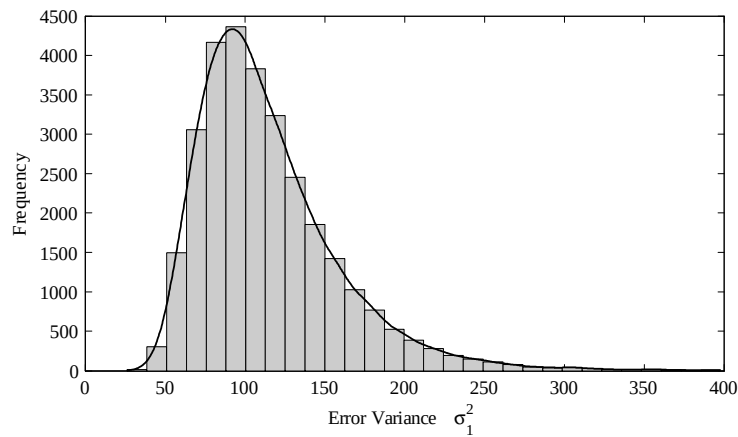


Figure 5.1. Histogram of σ_1^2 (error variance)

Mode: 107.17

95% Credibility Interval: [55.96 229.07]

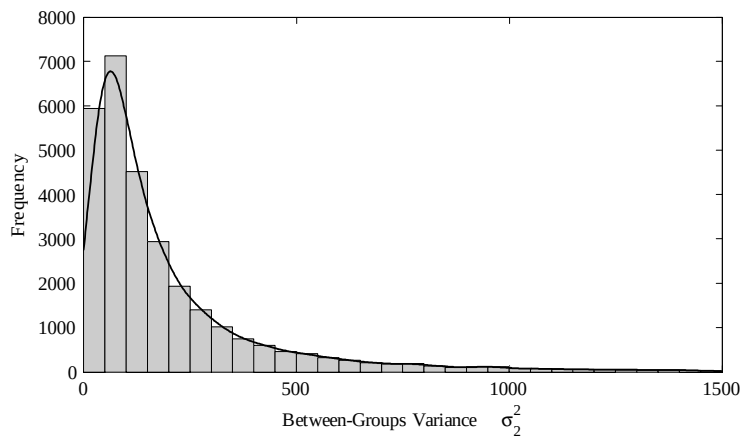


Figure 5.2. Histogram of σ_2^2 (between-groups variance)

Mode: 65.55

95% Credibility Interval: [12.76 850.87]

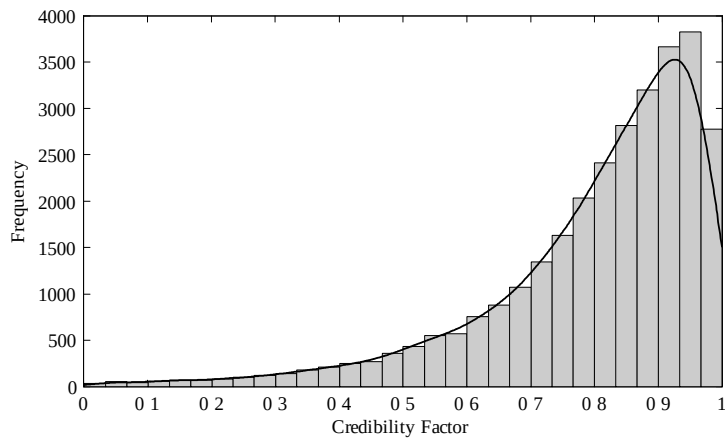


Figure 5.3. Histogram of $\tilde{Z}$ (credibility factor)

Mode: 0.83915

95% Credibility Interval: [0.28364 0.94541]

From equation (2.1), it follows that the credibility estimator for the i^{th} country is

$$\tilde{Z}\bar{Y}_{i.} + (1 - \tilde{Z})\bar{Y}_{..}$$

The point estimate of $\tilde{Z}$ in Figure 5.1 is 0.83915. From this it is clear that the best way to estimate or predict aggregate claim amounts for the i^{th} country is to use 84% of the average value of that country, $\bar{Y}_{i.}$, and 16% of the overall mean, $\bar{Y}_{..}$. This allows the i^{th} country to use both its own experience and the collective experience as a source of information. As mentioned, by averaging the conditional densities, the unconditional posterior density of θ_1 and the predictive densities of $\tilde{Y}_{1,1}$ and Y_1^* can easily be obtained. The posterior density of $\theta_1 = \theta + r_1$ and the predictive density of a future observation (claim) $\tilde{Y}_{1,1}$ from the first group (country) are displayed in Figure 5.4.

The posterior distribution of θ_1 illustrates the expected (average) aggregate claim amount for Country 1 over a number of years. The predictive density of $\tilde{Y}_{1,1}$ on the other hand is an indication of the aggregate claim amount of Country 1 for the following year. From the figure it is evident that the means and modes of the two distributions are the same, but that the 90% predictive interval of $\tilde{Y}_{1,1}$ is much wider than the corresponding credibility interval for θ_1 . This illustrates the uncertainty that is associated with the prediction of a future value. This fact is also clear if we compare $Var(\theta_i | \underline{Y}, \sigma_1^2, \sigma_2^2)$ and $Var(\tilde{Y}_{i,1} | \underline{Y}, \sigma_1^2, \sigma_2^2)$; equations (3.7) and (3.13) for $q = 1$.

In Figure 5.5, the predictive density of a future observation (claim) from a new or unknown group (country) is displayed. The mean $E(Y_1^* | \underline{Y}) = \bar{Y}_{..}$ and the wide prediction interval is obvious, since the variance of Y_1^* is very large, see equation (3.15).

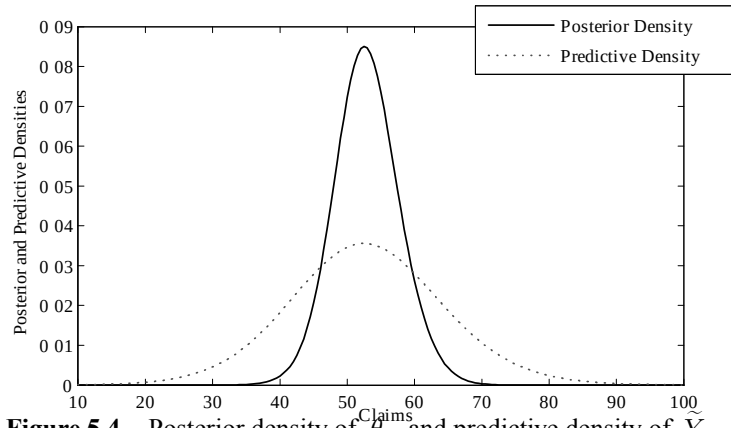


Figure 5.4. Posterior density of θ_1 and predictive density of $\tilde{Y}_{1,1}$

θ_1 : Mean = 54.44
 Mode = 52.90
 90% Credibility Interval: [45.75 61.55]
 $\tilde{Y}_{1,1}$: Mean = 54.13
 Mode = 52.90
 90% Credibility Interval: [34.88 73.22]

From Figure 5.5, it is clear that the best way of predicting the aggregate claim amount for a new, similar or unknown country for the following year, is to use the average value of the data. Since the effect or characteristic of this country is unknown, the predictive density and corresponding prediction interval will be fairly wide. The empirical Bayes estimate, $\hat{\hat{Z}}$, of the credibility factor, given on page 48 of Chapter 8 on Credibility Theory in the ActEd 106 Actuarial Study Guide, 2002, was calculated in the following way. Since $E(m_1) = \sigma_1^2$, an estimate of σ_1^2 is given by

$$\hat{\sigma}_1^2 = \frac{1}{\nu_1} \sum_{i=1}^I \sum_{j=1}^J (Y_{ij} - \bar{Y}_{ij.})^2 = \frac{1}{16} (1619.2) = 101.2.$$

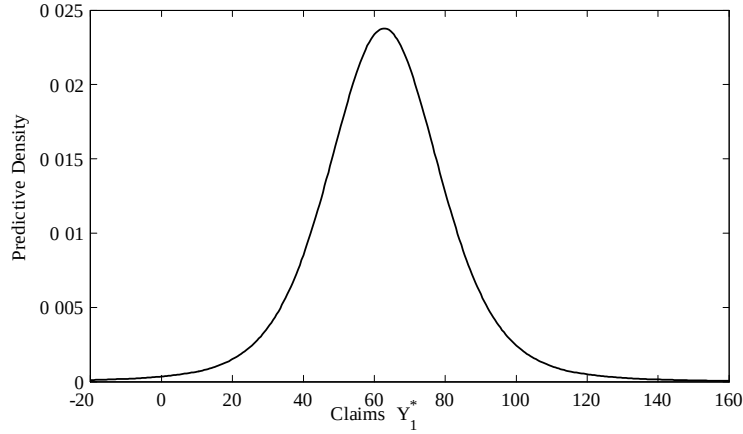


Figure 5.5. Predictive density of Y_1^*

Y_1^* : Mean = 62.75
 Mode = 62.75
 90% Credibility Interval: [36.65 88.85]

Also,

$$E(m_2) = \sigma_1^2 + J\sigma_2^2$$

and therefore

$$\hat{\sigma}_1^2 + J\hat{\sigma}_2^2 = \frac{1}{\nu_2} \sum_{i=1}^I (\bar{Y}_{i.} - \bar{Y}_{..})^2 = \frac{1}{3} (1658.55) = 552.85.$$

This means that $\hat{\sigma}_2^2 = 90.33$ and $\hat{\tilde{Z}} = \frac{J\hat{\sigma}_2^2}{\hat{\sigma}_1^2 + J\hat{\sigma}_2^2} = 0.8169$. If we compare these estimates to the point estimates given in Figures 5.1, 5.2 and 5.3, we see that the empirical Bayes estimates $\hat{\sigma}_1^2$ and $\hat{\tilde{Z}}$ are close to the point estimates for the full Bayesian analysis, illustrated in Figures 5.1 and 5.2. The point estimates of σ_2^2 on the other hand are quite different. As mentioned, the reason for this is the small number of degrees of freedom (countries) associated with σ_2^2 . In fact, the empirical Bayes estimate is $\hat{\sigma}_2^2 = 90.33$, while the mean and the mode for the full Bayesian analysis is 185.34 and 65.55 respectively, indicating

the skewness of the posterior distribution of σ_2^2 . In Table 5.2, estimates of the credibility premiums are calculated for the four countries by using a fully Bayesian analysis and compared to the empirical Bayes approach. The credibility intervals for θ_i ($i = 1, \dots, 4$) as well as the prediction intervals for future observations are also given.

Table 5.2. Means and credibility intervals for θ_i , as well as prediction intervals for future aggregate claim amounts

Country	Mean	90% Credibility Interval		90% Prediction Interval		Empirical Bayes Estimate
		Lower Limit	Upper Limit	Lower Limit	Upper Limit	
1	54.44	45.75	61.55	34.88	73.22	52.66
2	67.50	60.00	74.55	48.69	85.93	67.33
3	71.48	63.95	79.45	53.04	90.48	71.94
4	58.83	51.10	65.30	40.37	78.04	59.03

By comparing the two procedures, we see that the aggregate claim amounts which they yield are for all practical purposes the same. This is to be expected for it was mentioned by Harville (1974) that when “flat” or uniform priors are assigned to the fixed effects and non-informative priors to the variance components, the point estimates of the marginal posterior distributions will be close to the empirical Bayes estimates. The fully Bayesian approach has, however, some additional advantages:

- (i) The Bayesian practitioner does not need to commit himself to only a point estimate of the variance components and credibility estimator. Credibility intervals and predictive densities can easily be obtained.
- (ii) Uncertainty about the true values of the variance components is formally incorporated into the analysis through the choice of an appropriate prior distribution.
- (iii) The Bayesian approach provides a set of widely applicable and mathematically tractable tools, often more tailored to the requirement

of users than the corresponding frequentist tools.

One might also be interested in determining whether the average claim amounts of the four countries differ from one another, i.e. to test the hypothesis,

$$H_0 : \theta_1 = \theta_2 = \theta_3 = \theta_4 (= \theta_0) .$$

The classical approach to this is to obtain the likelihood ratio statistic. In the Bayesian case, the posterior distribution contains all the available information about a parameter. Therefore, one approach to determine if the average claim amounts of the four countries differ, is to calculate, for example, the 95% credibility intervals of the difference in parameter values for all possible pairs. If these intervals contain the value zero, the conclusion can be made that the hypothesis cannot be rejected. Another approach to test the same hypothesis is to calculate the Bayes factor, but we will only concentrate on the first approach. The following theorem is now needed.

Theorem 5.1 The posterior distribution of $\theta_l - \theta_m$ (the difference between two means), given the variance components, is normal with mean

$$E(\theta_l - \theta_m \mid \underline{Y}, \sigma_1^2, \sigma_2^2) = \frac{J\sigma_2^2}{\sigma_1^2 + J\sigma_2^2} (\bar{Y}_l - \bar{Y}_m.)$$

and variance

$$Var(\theta_l - \theta_m \mid \underline{Y}, \sigma_1^2, \sigma_2^2) = \frac{2\sigma_1^2\sigma_2^2}{\sigma_1^2 + J\sigma_2^2}.$$

The proof is given in Van der Merwe and Bekker (2004).

By using the simulation procedure as described in section 4, the posterior distribution of $\theta_1 - \theta_3$, i.e. the posterior distribution of the difference in average claim amounts between Country 1 and Country 3, is illustrated in Figure 5.6, and in Table 5.4 the 95% credibility intervals of the differences in claim amounts for all possible pairs are given.

Table 5.3. 95% Credibility intervals of the difference in claim amounts for all possible pairs

Pairs	95% Credibility Intervals of $\theta_l - \theta_m$		p-Values
	Lower Limit	Upper Limit	
(1,2)	-24.35	-2.70	0.0211
(1,3)	-29.10	-5.80	0.0040
(1,4)	-15.70	3.70	0.2842
(2,3)	-14.05	5.25	0.4379
(2,4)	-2.10	16.89	0.1666
(3,4)	1.67	20.54	0.0393

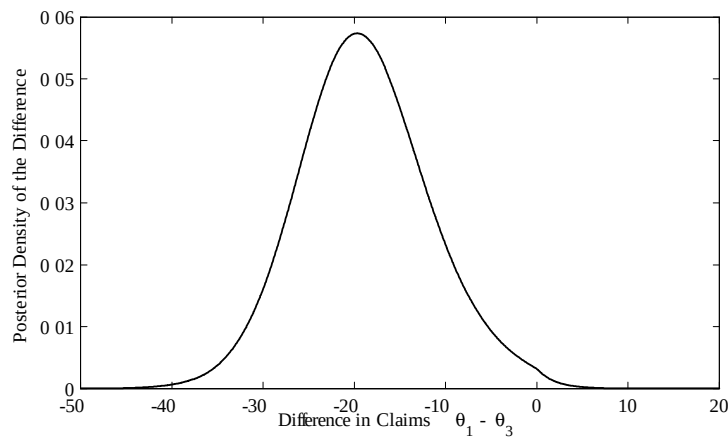


Figure 5.6. The posterior distribution of the difference in average claim amounts between Country 1 and Country 3

$\theta_1 - \theta_3$: Mean = -15.02
 Mode = -19
 95% Credibility Interval: [-29.1 -5.8]

From Table 5.3, as well as Figure 5.6, it is clear that the 95% interval for the pair (1,3) does not contain the value zero. This means that there is a significant difference in average claim amounts between Countries 1 and 3. This is also true for Countries 1 and 2, and Countries 3 and 4. Three of the intervals,

however, do contain zero, which is a good indicator of no real differences among these countries. We must, however, warn that indiscriminate use of this procedure may be dangerous, for if we have, say, I countries, where I is large, then the probability is high that at least one of the $\frac{1}{2}I(I-1)$ differences will, due to pure chance, be judged significantly different. In other words, this method will tend to give too many significantly different cases.

In the last column of Table 5.4 the p-values from the corresponding Likelihood Ratio tests are given. A comparison of the two procedures shows that the same conclusions will be made. Since the p-values for countries 1 and 2, 1 and 3, and 3 and 4 are smaller than 0.05, we conclude that there are real differences between these countries.

In the next section, it will be shown how Monte Carlo simulation and sampling importance resampling can be used to obtain independent draws from posterior distributions if proper priors for the variance components are used.

6. Sampling-importance resampling

The Sampling-Importance Resampling (SIR) algorithm aims at drawing a random sample from a target distribution π . First, a sample is drawn from a proposal distribution q , and then from this sample a smaller sample is drawn with sample probabilities proportional to the importance ratios $\frac{\pi}{q}$. For the algorithm to be efficient, it is important that q is a good approximation to π . This means that q should not have too light tails when compared to π . For further details, see Skare et al. (2003).

Let us assume for the moment that the prior distribution on the variance components σ_1^2 and σ_2^2 for the balanced Bühlmann model is such that the

prior distribution on the transformed parameters σ_1^2 and σ_{12}^2 is

$$q = p(\sigma_1^2, \sigma_{12}^2) \propto (\sigma_1^2)^{-\frac{1}{2}(\nu_0+2)} \exp\left\{-\frac{1}{2} \frac{s_1^2}{\sigma_1^2}\right\} (\sigma_{12}^2)^{-\frac{1}{2}(\nu_0+2)} \exp\left\{-\frac{1}{2} \frac{s_2^2}{\sigma_{12}^2}\right\} \quad (6.1)$$

where $\sigma_{12}^2 = \sigma_1^2 + J\sigma_2^2$ as before.

As mentioned by Guttman and Menzefricke (2003), the prior distribution on each of σ_1^2 and σ_{12}^2 is proportional to an inverse gamma distribution, but the restricted parameter space ($\sigma_{12}^2 > \sigma_1^2$) must be taken into account. The parameter ν_0 is common to both distributions, but this is done for notational convenience only. Using the prior distribution as defined in (6.1), we find that, if the restriction did not apply, the posterior distribution for σ_1^2 and σ_{12}^2 would be independent, each proportional to an inverse gamma distribution, i.e.

$$p(\sigma_1^2 | \underline{Y}) \propto (\sigma_1^2)^{-\frac{1}{2}(\nu_0+\nu_1+2)} \exp\left\{-\frac{1}{2} \frac{s_1^2 + \nu_1 m_1}{\sigma_1^2}\right\} \quad (6.2)$$

and

$$p(\sigma_{12}^2 | \underline{Y}) \propto (\sigma_{12}^2)^{-\frac{1}{2}(\nu_0+\nu_2+2)} \exp\left\{-\frac{1}{2} \frac{s_2^2 + \nu_2 m_2}{\sigma_{12}^2}\right\}. \quad (6.3)$$

But the restrictions do apply. However, by using the two-step rejection sampling procedure, as described in section 4, σ_1^2 and σ_2^2 can easily be simulated from the joint posterior distribution. Let $\xi = (\sigma_1^2, \sigma_2^2)$ and denote the $\tilde{l}$ independent draws of ξ by $\xi_1, \dots, \xi_{\tilde{l}}$. The posterior distribution, from which these sets of values were drawn, depends on the prior distribution q as defined in (6.1). Suppose now a more realistic (“correct”) prior distribution,

$$\pi = p(\sigma_1^2, \sigma_2^2) \propto (\sigma_1^2)^{-\frac{1}{2}(\nu_0+2)} \exp\left\{-\frac{1}{2} \frac{s_1^2}{\sigma_1^2}\right\} (\sigma_2^2)^{-\frac{1}{2}(\nu_0+2)} \exp\left\{-\frac{1}{2} \frac{s_2^2}{\sigma_2^2}\right\}. \quad (6.4)$$

According to Smith and Gelfand (1992), Guttman and Menzefrieke (2003) and Skare et al. (2003), the $\tilde{l}$ independent draws $\xi_1, \dots, \xi_{\tilde{l}}$ are a weighted sample from the posterior distribution based on π , where the weights are

$$w_i = \frac{\pi(\xi_i)/q(\xi_i)}{\sum_{i=1}^{\tilde{l}} \pi(\xi_i)/q(\xi_i)} \quad (6.5)$$

where $\pi(\xi_i)$ and $q(\xi_i)$ denote the realistic prior density and implied prior density defined in equations (6.1) and (6.4) respectively.

Using SIR, we have obtained independent samples from the “correct” joint posterior distribution. As mentioned by Guttman and Menzefrieke (2003), this method is preferable to methods such as Gibbs sampling where the latter generates dependent samples from the joint posterior distributions.

For illustrative purposes, let us now apply the results to Example 5.1. The quantities needed for simulation of the “approximate” posterior distribution, equations (6.2) and (6.3), are given in section 5. Since the degrees of freedom for some of the parameters are small ($\nu_2 = I - 1 = 3$), the posterior distributions are likely to be quite sensitive to the choice of prior parameters. In particular, suppose that the prior distribution for the transformed parameters, equation (6.1), are $\nu_0 = 1$ and $s_1^2 = s_2^2 = 0.00001$. In generating values from the “approximate” posterior distribution, equations (6.2) and (6.3), we simulated 33 490 in order to retain 30 000 sets of permissible values that met the restriction $\sigma_{12}^2 > \sigma_1^2$.

The purpose of this section, however, is to get independent draws from the “correct” posterior based on the prior π , defined in equation (6.4). For $\nu_0 = 1$ and $s_1^2 = s_2^2 = 0.00001$, the distribution of weights is given in Figure 6.1.

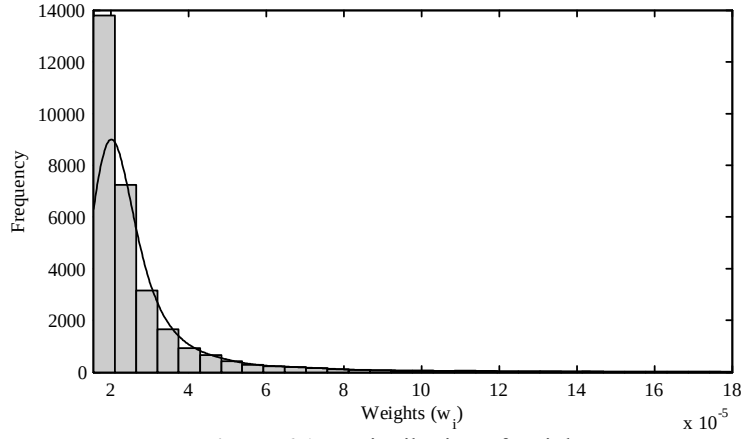


Figure 6.1. Distribution of weights

Using these weights, posterior distributions of σ_1^2 , σ_2^2 , the credibility factor $\tilde{Z}$, θ_i and the predictive distributions, all based on the prior π , can be obtained by drawing a smaller sample with replacement from the proposal distribution q with sample probabilities w_i , proportional to the importance ratio $\frac{\pi(\xi_i)}{q(\xi_i)}$. For this example the smaller sample consists of 10 000 observations.

In Figures 6.2 and 6.3 histograms of the posterior distributions of the variance components σ_1^2 and σ_2^2 based on the “correct” prior (equation (6.4)) are illustrated, and in Figure 6.4 the posterior distribution of the credibility factor $\tilde{Z} = \frac{J\sigma_2^2}{\sigma_1^2 + J\sigma_2^2}$ is given.

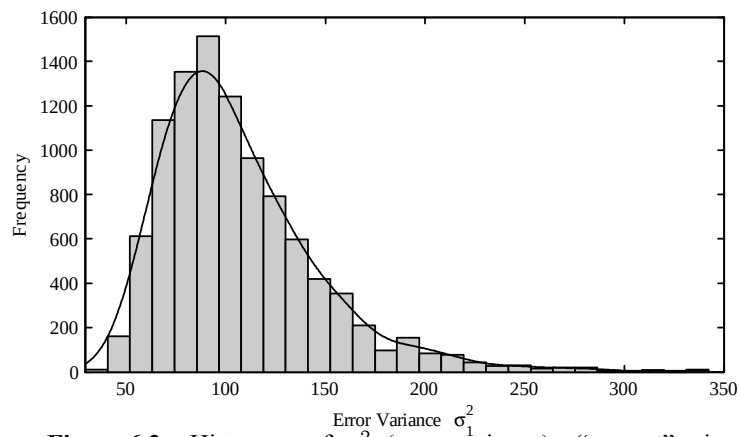


Figure 6.2. Histogram of σ_1^2 (error variance) – “correct” prior

Point Estimate: 98.20

95% Credibility Interval: [57.85 203.07]

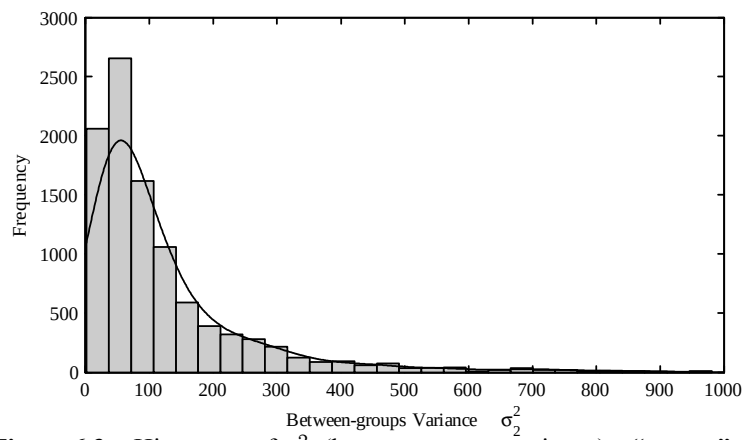


Figure 6.3. Histogram of σ_2^2 (between-groups variance) – “correct” prior

Point Estimate: 68.49

95% Credibility Interval: [14.55 537.42]

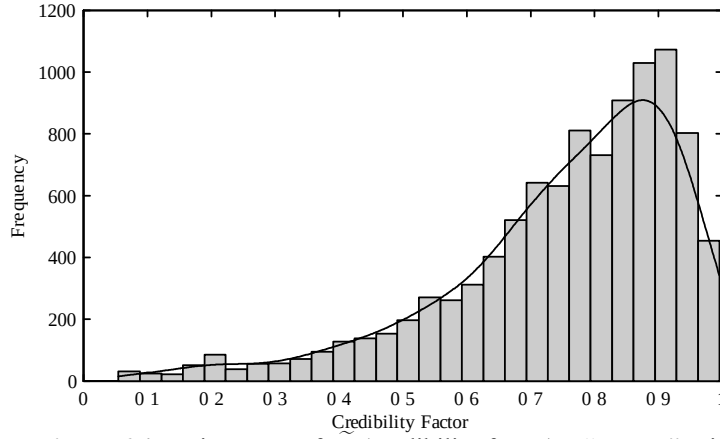


Figure 6.4. Histogram of $\tilde{Z}$ (credibility factor) – “correct” prior

Point Estimate: 0.87167

95% Credibility Interval: [0.31506 0.94727]

The point estimate of the credibility factor $\tilde{Z} = 0.87167$ in the case of the proper (“correct”) prior is somewhat larger than the point estimate $\tilde{Z} = 0.83915$ for the noninformative prior defined in equation (3.2). This difference is mainly attributed to the smaller error variance which is the result of the more informative (proper) prior on σ_1^2 . In Table 6.1 a comparison is made between the variance components and credibility factors for two different posterior distributions. The first posterior is obtained by using the improper prior $p(\sigma_1^2, \sigma_2^2) \propto \sigma_1^{-2} (\sigma_1^2 + J\sigma_2^2)^{-1}$ and the second from the “correct” and proper prior $p(\sigma_1^2, \sigma_2^2) \propto \prod_{i=1}^2 (\sigma_i^2)^{-\frac{1}{2}(\nu_0+2)} \exp\left\{-\frac{1}{2}\frac{s_i^2}{\sigma_i^2}\right\}$. The modes and the 95% credibility (Bayes confidence) intervals are given.

Table 6.1. A comparison between the variance components and credibility factors for posteriors using different priors

	Improper Prior: $\sigma_1^{-2}(\sigma_1^2 + \mathbf{J}\sigma_2^2)^{-1}$			Proper Prior: $\prod_{i=1}^2 (\sigma_i^2)^{-\frac{1}{2}(\nu_0+2)} \exp\left\{-\frac{1}{2} \frac{s_i^2}{\sigma_i^2}\right\}$		
	σ_1^2	σ_2^2	Credibility Factor	σ_1^2	σ_2^2	Credibility Factor
Mode	107.17	65.55	0.8392	98.20	70.96	0.8717
Lower 95% Limit	55.96	12.76	0.2836	57.85	14.55	0.3151
Upper 95% Limit	229.07	850.87	0.9454	203.07	537.42	0.9473

It is evident from Table 6.4 that the point estimates of the two posteriors are quite close to one another but that the credibility intervals differ quite a lot. This phenomenon is the largest for σ_2^2 where the upper 95% credibility limit, in the case of the improper prior, is 850.87, which is much larger than the corresponding limit of 437.59 for the proper prior.

This is not surprising, because proper priors contain more information (more degrees of freedom), i.e. less uncertainty in the posteriors and will result in smaller credibility intervals for the unknown parameters.

The assignment of proper priors to specific parameters, however, must be justifiable from a practical point of view. It is further clear from the figures that the posterior densities are quite skew (except for σ_1^2), which means that the means, modes and medians of the distributions will not coincide.

A further important aspect is that by using the weights in (6.5), estimates of functions of ξ , based on the “correct” posterior, can readily be obtained. For example, an estimate of the posterior mean of the function $f(\xi)$ is

$$\hat{E}(f) = \sum_{i=1}^{\bar{I}} w_i f(\xi_i).$$

Posterior variances, third and fourth moments about the mean, can similarly be obtained and used to calculate Pearson curve approximations of the posterior distributions. Estimated posterior densities of π can therefore be obtained without resampling from the proposal distribution q .

Klugman (1992) concludes that it is of growing importance to the actuary that not just the expected future outcomes and parameter estimates are taken into account, but the variability of the predictions and the posterior distributions of the parameters. In measuring variability it is essential that as many sources as possible should be taken into account. The two most obvious sources are the natural process variability that is inherent in the model and the variability because the model is estimated from data. From our results, it is clear that the Bayesian approach provides an ideal mechanism for combining these two sources of variability.

7. Conclusion

In this note a Bayesian analysis to develop credibility estimates of the balanced Bühlmann model is proposed. Simple numerical methods to obtain exact posterior distributions and predictive densities are described. These distributions are obtained through Monte Carlo simulations that generate independent samples from the joint posterior distribution. The method is therefore easier to use than methods such as Gibbs sampling, which generates dependent samples from the joint distribution.

A comparison between the fully Bayesian analysis and the empirical Bayes method, in the case of a non-informative prior, shows that there is not much to choose between the two procedures. The fully Bayesian approach has, however, some additional advantages. Credibility intervals and predictive densities can easily be obtained, and a set of widely applicable and

mathematically tractable tools are provided, which are often more tailored to the requirement of users than the corresponding frequentist tools.

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APPENDIX A

Proof of Theorem 3.1

First, we will briefly review the theory for the probability matching prior. Datta and Ghosh (1995) derived the differential equation that a prior must satisfy if the posterior probability of a one-sided credibility interval for a parametric function and its frequentist probability agree up to $O(n^{-1})$ where n is the sample size. They proved that the agreement between the posterior probability and the frequentist probability holds if and only if

$$\sum_{\alpha=1}^m \frac{\partial}{\partial \theta_{\alpha}} \left\{ \eta_{\alpha}(\underline{\theta}) \pi(\underline{\theta}) \right\} = 0$$

where $\pi(\underline{\theta})$ is the probability matching prior for $\underline{\theta}$, the vector of unknown parameters.

Also,

$$\nabla_t(\underline{\theta}) = \left[\frac{\partial}{\partial \theta_1} t(\underline{\theta}) \dots \frac{\partial}{\partial \theta_m} t(\underline{\theta}) \right]'$$

and

$$\eta(\underline{\theta}) = \frac{F^{-1}(\underline{\theta}) \nabla_t(\underline{\theta})}{\sqrt{\nabla_t'(\underline{\theta}) F^{-1}(\underline{\theta}) \nabla_t(\underline{\theta})}} = [\eta_1(\underline{\theta}) \dots \eta_m(\underline{\theta})]'$$

It is clear that $\eta'(\underline{\theta}) F^{-1}(\underline{\theta}) \eta(\underline{\theta}) = 1$ for all $\underline{\theta}$, where $F^{-1}(\underline{\theta})$ is the inverse of $F(\underline{\theta})$, the Fisher Information matrix of $\underline{\theta}$, and $t(\underline{\theta})$ is the parameter of interest.

The Fisher Information matrix is given by

$$F(\theta, \sigma_2^2, \sigma_1^2) = F(\tilde{\theta}) = \begin{bmatrix} \frac{IJ}{\sigma_1^2 + J\sigma_2^2} & 0 & 0 \\ 0 & \frac{IJ^2}{2(\sigma_1^2 + J\sigma_2^2)^2} & \frac{IJ}{2(\sigma_1^2 + J\sigma_2^2)^2} \\ 0 & \frac{IJ}{2(\sigma_1^2 + J\sigma_2^2)^2} & \frac{I(J-1)}{2(\sigma_1^2)^2} + \frac{I}{2(\sigma_1^2 + J\sigma_2^2)^2} \end{bmatrix}$$

(see Box & Tiao (1973), Chapter 5, page 252); and the inverse of the Fisher Information Matrix by

$$F^{-1}(\theta, \sigma_2^2, \sigma_1^2) = F^{-1}(\tilde{\theta})$$

$$= \begin{bmatrix} \frac{\sigma_1^2 + J\sigma_2^2}{IJ} & 0 & 0 \\ 0 & \frac{\left\{ \frac{I(J-1)}{2(\sigma_1^2)^2} + \frac{I}{2(\sigma_1^2 + J\sigma_2^2)^2} \right\}}{|H|} & \frac{\frac{-IJ}{2(\sigma_1^2 + J\sigma_2^2)^2}}{|H|} \\ 0 & \frac{\frac{-IJ}{2(\sigma_1^2 + J\sigma_2^2)^2}}{|H|} & \frac{\frac{IJ^2}{2(\sigma_1^2 + J\sigma_2^2)^2}}{|H|} \end{bmatrix}.$$

The determinant $|H|$ is equal to

$$\begin{aligned} |H| &= \frac{IJ}{2(\sigma_1^2 + J\sigma_2^2)^2} \left\{ \frac{I(J-1)}{2(\sigma_1^2)^2} + \frac{I}{2(\sigma_1^2 + J\sigma_2^2)^2} \right\} - \frac{I^2 J^2}{4(\sigma_1^2 + J\sigma_2^2)^4} \\ &= \frac{I^2 J^2}{4(\sigma_1^2 + J\sigma_2^2)^2} \left\{ \frac{(J-1)}{(\sigma_1^2)^2} \right\}. \end{aligned}$$

We are interested in the probability matching prior for σ_1^2 , the residual (error) variance.

Let $\tilde{\theta} = [\theta, \sigma_2^2, \sigma_1^2]'$ and $t(\tilde{\theta}) = \sigma_1^2$, then

$$\frac{\partial t(\tilde{\theta})}{\partial \theta} = 0; \quad \frac{\partial t(\tilde{\theta})}{\partial \sigma_2^2} = 0; \quad \frac{\partial t(\tilde{\theta})}{\partial \sigma_1^2} = 1; \quad \nabla'_t(\tilde{\theta}) = [0 \ 0 \ 1]$$

and

$$\nabla'_t(\tilde{\theta}) F^{-1}(\tilde{\theta}) = \begin{bmatrix} 0 & \frac{-IJ}{\frac{2(\sigma_1^2 + J\sigma_2^2)^2}{|H|}} & \frac{IJ^2}{\frac{2(\sigma_1^2 + J\sigma_2^2)^2}{|H|}} \end{bmatrix}.$$

Further

$$\sqrt{\nabla'_t(\tilde{\theta}) F^{-1}(\tilde{\theta}) \nabla_t(\tilde{\theta})} = \sqrt{\frac{IJ^2}{\frac{2(\sigma_1^2 + J\sigma_2^2)^2}{|H|}}}$$

and

$$\begin{aligned} \frac{\nabla'_t(\tilde{\theta}) F^{-1}(\tilde{\theta})}{\sqrt{\nabla'_t(\tilde{\theta}) F^{-1}(\tilde{\theta}) \nabla_t(\tilde{\theta})}} &= \begin{bmatrix} 0 & \frac{-I^{\frac{1}{2}}}{\sqrt{2(\sigma_1^2 + J\sigma_2^2)} \sqrt{|H|}} & \frac{I^{\frac{1}{2}} J}{\sqrt{2(\sigma_1^2 + J\sigma_2^2)} \sqrt{|H|}} \end{bmatrix} \\ &= \begin{bmatrix} \eta_1(\tilde{\theta}) & \eta_2(\tilde{\theta}) & \eta_3(\tilde{\theta}) \end{bmatrix}. \end{aligned}$$

The prior $p(\tilde{\theta}) = p(\theta, \sigma_2^2, \sigma_1^2)$ is a probability matching prior if the following differential equation is satisfied:

$$\frac{\partial \left\{ \eta_1(\tilde{\theta}) p(\tilde{\theta}) \right\}}{\partial \theta} + \frac{\partial \left\{ \eta_2(\tilde{\theta}) p(\tilde{\theta}) \right\}}{\partial \sigma_2^2} + \frac{\partial \left\{ \eta_3(\tilde{\theta}) p(\tilde{\theta}) \right\}}{\partial \sigma_1^2} = 0.$$

Take $p(\tilde{\theta}) = \sqrt{|H|}$, then

$$(i) \quad \frac{\partial \left\{ \eta_1(\tilde{\theta}) p(\tilde{\theta}) \right\}}{\partial \theta} = 0,$$

$$(ii) \quad \frac{\partial \left\{ \eta_2 \left(\tilde{\theta} \right) p \left(\tilde{\theta} \right) \right\}}{\partial \sigma_2^2} = \frac{\partial}{\partial \sigma_2^2} \left\{ \frac{-I^{\frac{1}{2}}}{\sqrt{2} (\sigma_1^2 + J\sigma_2^2)} \right\} = \frac{I^{\frac{1}{2}} J}{\sqrt{2} (\sigma_1^2 + J\sigma_2^2)^2}$$

$$(iii) \quad \frac{\partial \left\{ \eta_3 \left(\tilde{\theta} \right) p \left(\tilde{\theta} \right) \right\}}{\partial \sigma_1^2} = \frac{\partial}{\partial \sigma_1^2} \left\{ \frac{I^{\frac{1}{2}} J}{\sqrt{2} (\sigma_1^2 + J\sigma_2^2)} \right\} = \frac{-I^{\frac{1}{2}} J}{\sqrt{2} (\sigma_1^2 + J\sigma_2^2)^2}.$$

It therefore follows that

$$\frac{\partial \left\{ \eta_1 \left(\tilde{\theta} \right) p \left(\tilde{\theta} \right) \right\}}{\partial \theta} + \frac{\partial \left\{ \eta_2 \left(\tilde{\theta} \right) p \left(\tilde{\theta} \right) \right\}}{\partial \sigma_2^2} + \frac{\partial \left\{ \eta_3 \left(\tilde{\theta} \right) p \left(\tilde{\theta} \right) \right\}}{\partial \sigma_1^2} = 0$$

and

$$\begin{aligned} p \left(\tilde{\theta} \right) &= p \left(\theta, \sigma_2^2, \sigma_1^2 \right) = \sqrt{|H|} = \frac{IJ}{2 (\sigma_1^2 + J\sigma_2^2)} \frac{\sqrt{J-1}}{\sigma_1^2} \\ &\propto \frac{1}{\sigma_1^2 (\sigma_1^2 + J\sigma_2^2)} = \sigma_1^{-2} (\sigma_1^2 + J\sigma_2^2)^{-1}. \end{aligned}$$